

Raghav Ramani

GitHub | LinkedIn | Portfolio | raghavramani2004@gmail.com | +91 8003708227 | Jaipur, India

SUMMARY

Full-stack AI/ML engineer specializing in multi-agent orchestration, production RAG pipelines, and full-stack GenAI applications. Builds enterprise and GovTech platforms end-to-end — vector ingestion, LangGraph execution, FastAPI backends, Next.js UIs, live database integrations.

WORK EXPERIENCE

Aasvaa Innovation Labs, Jaipur — Full-Stack + AI Engineer Intern Jan–Jul 2026

- Co-developed the GovTech Planning Intelligence Platform ("Ask Manthaa OS") – full-stack features + AI chatbot layer over live Supabase data across 41 districts (see Projects).
- Architected a 7-stage multi-agent Legal AI engine (FastAPI + LangGraph + Groq Llama 3.1 70B) with hybrid retrieval (Tavily real-time search + Qdrant over 70K+ Supreme Court vectors) – 3.2% hallucination rate, <2.2s latency.

Klear AI — AI/ML Engineer Intern Aug–Nov 2025

- Built production RAG pipelines (LangChain + Qdrant + Groq Llama 3.1 70B) with embedding optimization and custom retrieval tuning.
- Developed GenAI workflows: prompt-engineering pipelines, CNN image-detection models, evaluation benchmarks.

Accio Job — Teaching Assistant (Java) Aug 2023 – Feb 2024

- Co-delivered Java + OOP curriculum to 50 students; led code reviews, doubt sessions, technical workshops.

PROJECTS

GovTech Planning Intelligence Platform – Live

- District-development planning platform aggregating governance data across 41 districts, 457 blocks, 14,000+ local bodies for a state government initiative.
- Built "Ask Manthaa OS": full-stack Hindi/English NLP assistant over 11.87L aspirations, automating district-level strategic briefs (42 PDFs) and 9-sector KPI dashboards at 95.87% uptime.
- Engineered an aspiration-driven budget allocation engine computing optimal fund distribution by criticality score, baseline deficit, and funding status per sector.

SarkariSaathi — AI GovTech Platform

- AI-powered GovTech platform helping Indian citizens navigate government services (Ration Card, Aadhaar, PAN) across 30 states without paid middlemen.
- Built on Next.js 14 + Supabase with full Hindi/English i18n, seeding 27+ state-specific guides via a single SQL migration; Gemini-powered intent detection, BetterAuth (Google OAuth), ArcJet-secured rate limiting.
- Cut query-to-answer time from 15–20s (manual portal navigation) to 2–3s (AI-routed guide) – roughly 80–85% reduction; SEO-optimized structured metadata and semantic HTML achieved full search-engine indexing (3/3 pages) shortly after launch.

Legal AI System – GitHub

- 7-stage deterministic multi-agent research engine (LangGraph + Groq Llama 3.1 70B) over 70K+ Indian Supreme Court judgments.
- Hybrid retrieval: live web research via Tavily API fused with semantic vector search via Qdrant.
- Confidence-scored outputs [30–95], self-healing JSON parsing, hallucination guard – 0.91 retrieval precision, 3.2% hallucination rate.

ArthMitra — AI-Powered Financial Security Platform – GitHub

- LangGraph Supervisor routing a 3-agent swarm (Auditor/Shield/Mitra) across Deepseek-R1, Qwen2.5, Gemma3, Groq Llama-3.1.
- Shield ML: TF-IDF+LR, Random Forest, One-Class SVM for real-time SMS-scam detection, transaction-risk scoring, credit profiling; Next.js glassmorphism UI.

SKILLS

Languages:	Python, SQL, JavaScript (intermediate)
Frontend:	Next.js, React, Streamlit, HTML/CSS
Backend/DB:	FastAPI, Flask, Async Python, Supabase, PostgreSQL, REST APIs
AI/GenAI:	LangChain, LangGraph, MCP, RAG, Multi-Agent Systems, Prompt Engineering, Groq, Gemini, Ollama
Vector/NLP:	Qdrant, ChromaDB, FAISS, Sentence Transformers, spaCy, NLTK

EDUCATION

JECRC University 2022 – 2026

B.Tech CSE (AI & ML) – Specialization: Artificial Intelligence & Machine Learning | **CGPA: 7.2**

CERTIFICATIONS & ACHIEVEMENTS

- **LangChain Academy:** 9 certifications – LangChain, LangGraph, LangSmith, agentic architectures.
- **SQL for Data Analysis:** Intellipaat (joins, window functions, subqueries, aggregations).
- **GSEA Winner:** Suraksha Nivesh – AI stock-scam prediction project.